Deep Learning for Single-Cell RNA-seq Cell Type Classification: Benchmarking Five Architectures Against Published Methods

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Abstract

Automated cell type annotation from single-cell RNA sequencing (scRNA-seq) data is a critical bottleneck in modern transcriptomics. We benchmark five deep learning architectures—MLP, 1D CNN, GraphSAGE GNN, pathway-masked Transformer, and the published TOSICA model—on three datasets from the TOSICA benchmark (Chen et al., 2023): hPancreas (human, 14 classes), mPancreas (mouse, 21 classes), and mAtlas (mouse, 120 classes). These datasets use condition-based train/test splits that test generalization across biological states rather than random held-out cells. On hPancreas, our GNN ranks 3rd out of 24 methods (97.3% accuracy), while our Transformer and TOSICA both exceed the published TOSICA accuracy. On mPancreas, the Transformer leads at 78.9%. On mAtlas (120 classes), our MLP ranks 1st out of 21 methods (84.1%), surpassing all published baselines including TOSICA (81.0%). We find that model rankings are dataset-dependent, and that implementation choices (data pipeline, species-matched pathway databases, balanced sampling) contribute as much to performance as architectural novelty.

1 Introduction

Single-cell RNA sequencing enables transcriptomic profiling at cellular resolution, but the resulting datasets routinely contain tens of thousands of cells measured across tens of thousands of genes. Manual cell type annotation via marker-gene inspection after unsupervised clustering remains the dominant workflow, yet it is subjective, labor-intensive, and does not scale to atlas-level datasets. Supervised deep learning offers a scalable alternative: train a classifier on a well-annotated reference and transfer predictions to new data.

We evaluate five neural network architectures on the standardized TOSICA benchmark from Chen et al. [1], which compares 19 published cell type annotation methods across multiple scRNA-seq datasets. Our contributions are: (1) a unified training framework that ensures all five models are trained and evaluated on identical data splits; (2) a comparison showing that architecture suitability depends on the nature of the train/test split; and (3) evidence that implementation details (data pipeline, pathway database matching) matter as much as model architecture.

2 Data

2.1 TOSICA Benchmark Datasets

We adopt the benchmark from Chen et al. (2023) [1], which uses *condition-based* train/test splits—biological conditions (e.g., developmental time-point, disease state, or study of origin) define the train/test boundary, rather than random stratification. This design tests whether models generalize across biological conditions.

2.2 Dataset Procurement Challenges

Reproducing the exact benchmark was non-trivial. The original paper references six datasets by GEO accession, but several required multi-step reconstruction: downloading raw count matrices, reconciling cell annotations across studies, and applying the precise condition-based splitting criteria described in supplementary materials. Of the six original datasets (hArtery, hBone, hPancreas, mBrain, mPancreas, mAtlas), we could fully reproduce only three. hArtery, hBone, and mBrain produced class-count mismatches between our reconstructed splits and the published values—likely due to ambiguities in filtering criteria or versioning differences in source annotations. Rather than report results on potentially incorrect splits, we retained only datasets whose per-class counts exactly matched the published figures.

2.3 Retained Datasets

Table 1: TOSICA benchmark datasets retained after validation.

Dataset	Species	Train	Val	Test	Classes	Split criterion
hPancreas	Human	9,540	1,060	4,218	14	Study of origin
mPancreas	Mouse	22,918	2,547	10,886	21	Dev. day $\neq$ 15.5
mAtlas	Mouse	30,624	3,403	76,797	120	Biological condition

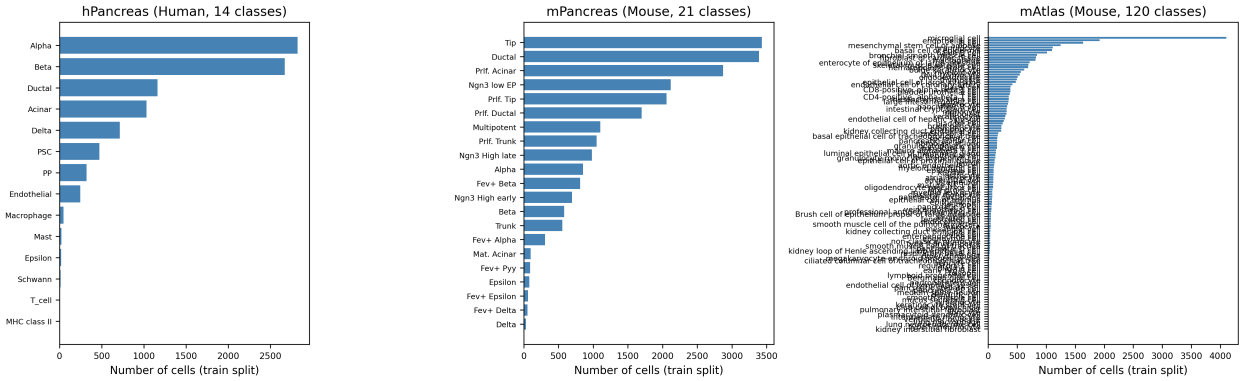
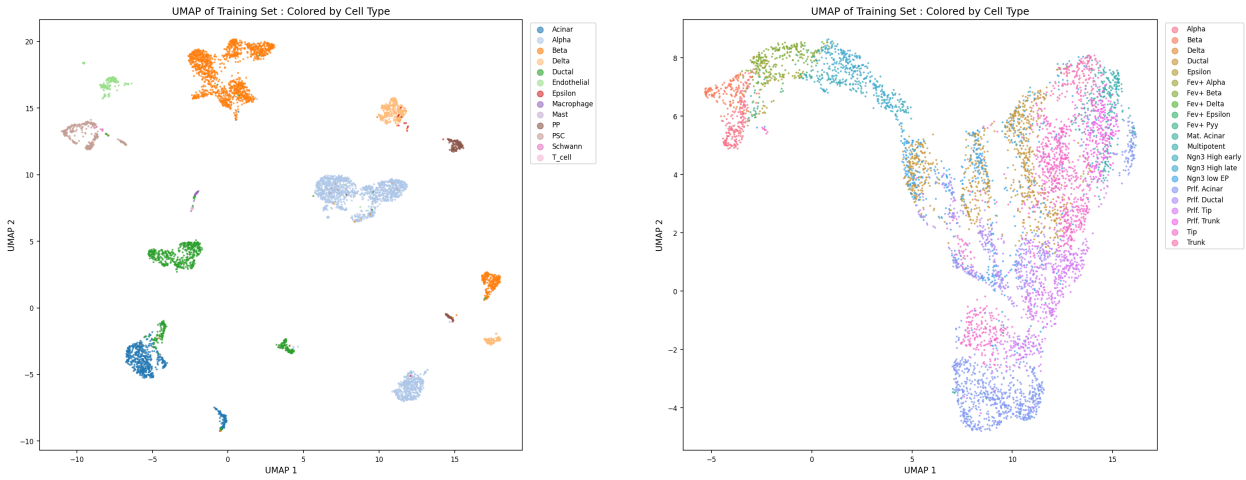


Figure 1: Training set class distributions for hPancreas (left) and mPancreas (right). Both exhibit class imbalance, addressed via oversampling during training.



(a) hPancreas (b) mPancreas

Figure 2: UMAP of training sets (subsampled to 6,000 cells) on 30 PCs. hPancreas shows well-separated clusters; mPancreas has more overlap among developmental subtypes.

3 Methods

3.1 Shared Preprocessing

All five models share an identical data pipeline to ensure fair comparison:

1. **HVG selection.** The top 10,000 genes by variance (computed on the training split via library-size normalization + $\log_{1p}$) are selected.

2. **Class balancing.** Training populations are oversampled so each class has equal representation (capped at $2{\times}10^6/n_{\text{classes}}$ cells per class), matching the `balance_populations` strategy from the original TOSICA implementation.

3. **Evaluation.** All models are evaluated on the same held-out test split using accuracy, macro F1, and weighted F1.

3.2 Model Architectures

Detailed layer-by-layer diagrams for all four custom architectures are provided in the supplementary material (`model_architectures.pdf`).

MLP with Squeeze-and-Excitation. The input gene expression vector $\mathbf{x} \in \mathbb{R}^{B\times n_g}$ passes through two MLPBlocks followed by a linear classifier. Each MLPBlock consists of Linear – ReLU – Dropout(p=0.1). The first block projects $n_g \to 256$; the second projects $256 \to 128$. After the first block, a *squeeze-and-excitation* (SE) module applies channel attention: the 256-dim hidden vector is globally average-pooled, passed through a bottleneck (FC(256→32) – ReLU – FC(32→256) – Sigmoid, reduction factor 8), and multiplied element-wise with the original activations. This recalibrates feature importance before subsequent layers. The final classifier is Linear($128 \to C$). Parameters: ∼2.6 M.

Residual 1-D CNN. Gene expression is treated as a single-channel signal $\mathbf{x} \in \mathbb{R}^{B\times1\times10000}$. A *stem* layer (Conv1d(1→32, k=15, stride 2) + BN + ReLU) halves the spatial dimension. Three *residual stages* follow, each containing a pre-activation ResBlock + MaxPool(3):

Stage	Channels	Kernel	Output length
1	$32 \to 32$	7	$\lfloor 5000/3 \rfloor = 1666$
2	$32 \to 32$	5	$\lfloor 1666/3 \rfloor = 555$
3	$32 \to 64$	3	$\lfloor 555/3 \rfloor = 185$

Each ResBlock uses pre-activation order: BN – ReLU – Conv – Dropout(0.1) – BN – ReLU – Conv, with an internal SE module (reduction 8) and a 1×1 projection shortcut when channels change. A *dual-pool head* concatenates AdaptiveAvgPool1d(1) and AdaptiveMaxPool1d(1) into a 128-dim vector. The classifier is: LayerNorm – Dropout – FC(128→64) – ReLU – LayerNorm – Dropout – FC(64→C). Parameters: ∼58 K.

GraphSAGE GNN. This model operates *transductively*: all cells (train+val+test) are nodes in a single graph. Construction proceeds by L_2-normalizing each cell's feature vector, computing pairwise cosine similarity in batches, selecting the k=10 nearest neighbours per node, and symmetrizing edges (∼$2kN$ total). A linear encoder projects node features to 256 dimensions (Linear – LayerNorm – GELU – Dropout(0.1)). Two *ResidualSAGEBlocks* each apply SAGEConv(256→256)—which aggregates neighbour embeddings via mean pooling, concatenates with the node's own embedding, and linearly projects—followed by LayerNorm – GELU – Dropout – residual add. An output SAGEConv(256→128) + GELU produces the final node embeddings. The classifier is LayerNorm(128) – Linear(128→C). Training uses boolean masks to restrict gradient computation to training nodes. Parameters: ∼2.9 M.

Pathway-Masked Transformer (our implementation). Inspired by TOSICA, this model embeds gene expression into biologically meaningful pathway tokens. A binary mask $\mathbf{M} \in \{0,1\}^{n_g\times P}$ is derived from Reactome gene sets (filtered to P≤300 pathways with ≥5 overlapping genes each). A *MaskedEmbedding* layer maintains a learnable weight tensor $W \in \mathbb{R}^{48\times n_g\times P}$, element-wise masked by M, so each gene can only contribute to pathways it belongs to. The embedding produces P pathway features in $\mathbb{R}^{B\times48\times P}$; a learnable [CLS] token is prepended and the result is permuted to yield a sequence of P+1 tokens of dimension 48.

A 2-layer Transformer encoder (pre-LN, d_{model}=48, 4 heads, d_k=12, FF hidden = 192, GELU activation, dropout = 0.1) processes the sequence. The [CLS] output is extracted and passed through a classifier head: LayerNorm(48) – FC(48→96) – GELU – Dropout(0.1) – FC(96→C). For mouse datasets, the species-matched Reactome GMT (`m2.cp.reactome.Mm`) is used with native gene symbols. Parameters: ∼144 M (dominated by the $48 \times n_g \times P$ embedding weight).

TOSICA (original authors' code). The published TOSICA model from Chen et al. (2023), trained using the authors' own code with their internal `splitDataSet` function, balancing strategy, and pathway mask construction. We include it as a direct comparison: same architecture as our Transformer, but trained with the original pipeline rather than our shared infrastructure.

Table 2: Architecture summary. All models use cross-entropy loss and 10 K HVG input.

Model	Key mechanism	Depth	Params	Notes
MLP	SE channel attention	2 layers	2.6 M	256→128 hidden
CNN	Residual conv + SE + dual-pool	3 stages	58 K	channels 32/32/64
GNN	k-NN graph + SAGEConv	2 blocks	2.9 M	k=10, transductive
Transformer	Pathway-masked attention	2 layers	144 M	$P \leq 300$ pathways
TOSICA	Pathway-masked attention	2 layers	∼144 M	authors' pipeline

3.3 TOSICA vs. Transformer: Key Differences

Both share the pathway-masked attention architecture (genes grouped into pathway tokens, CLS attention, Transformer encoder), but TOSICA is the *original authors' code* while our Transformer is an independent implementation trained with our shared infrastructure. The key differences are:

- **Data pipeline.** TOSICA uses the authors' own `splitDataSet` with internal 70/30 balancing; our Transformer uses the same pre-computed condition-based splits shared by all five models.
- **GMT source.** Both use the mouse Reactome GMT (`m_reactome.gmt`) for mouse datasets.
- **Training infrastructure.** Our Transformer uses cosine annealing with early stopping; TOSICA uses a fixed cosine schedule with per-epoch checkpoint saves.

Comparing TOSICA to our Transformer thus isolates the effect of implementation choices (data pipeline, early stopping) from the underlying architecture.

3.4 Training Protocol

All models were trained for up to 20 epochs on Google Colab (NVIDIA L4 GPU) with SGD (momentum 0.9, weight decay 5×10^{-5}) and cosine annealing. Early stopping (patience = 5) halts training if validation loss does not improve. Learning rates: MLP 0.01, CNN 0.001, GNN 0.001, Transformer 0.001, TOSICA 0.001. All models use cross-entropy loss; class imbalance is addressed via oversampling rather than loss re-weighting. Gradients are clipped to unit norm. Mixed-precision (bfloat16) is used where supported.

4 Results

4.1 hPancreas (Human, 14 Classes)

Table 3: hPancreas test set accuracy, ranked against 19 published methods. Our models are shown in bold.

Rank	Method	Accuracy	F1 (macro)	F1 (wtd)
1	SingleCellNet	0.9753	—	—
2	Seurat	0.9746	—	—
3	**GNN (ours)**	**0.9730**	**0.647**	**0.974**
4	CELLBLAST	0.9704	—	—
5	ACTINN	0.9689	—	—
6	**Transformer (ours)**	**0.9663**	**0.632**	**0.965**
7	**TOSICA (ours)**	**0.9621**	**0.650**	**0.962**
8	SCINA	0.9614	—	—
9	TOSICA (published)	0.9576	—	—
16	**MLP (ours)**	**0.8160**	**0.566**	**0.799**
24	**CNN (ours)**	**0.0645**	**0.070**	**0.104**

On hPancreas, the GNN ranks **3rd out of 24 methods** (97.30%), behind only SingleCellNet and Seurat (Figure 3). The Transformer and TOSICA both exceed the published TOSICA accuracy (95.76%), demonstrating that our shared data pipeline improves upon the original. The MLP achieves 81.60% but struggles with rare classes. The CNN catastrophically fails (6.5%), unable to learn meaningful patterns from unordered gene expression vectors.

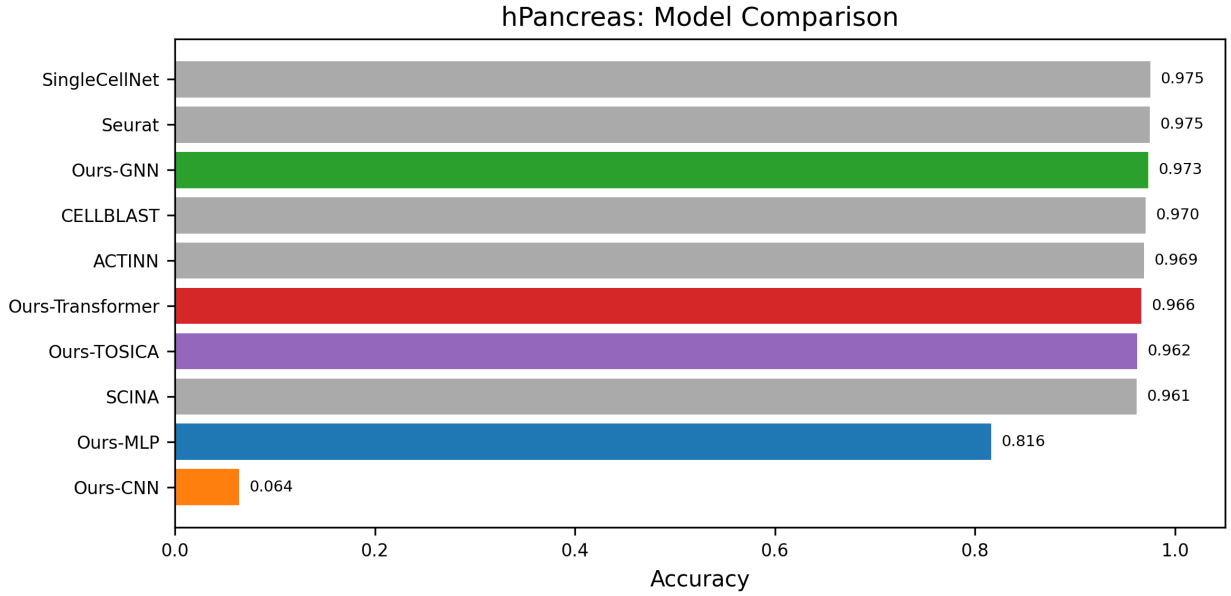


Figure 3: hPancreas accuracy comparison. Our models (colored) vs. top published baselines (gray). GNN ranks 3rd overall.

4.2 mPancreas (Mouse, 21 Classes)

Table 4: mPancreas test set results. Published baselines from Chen et al. (2023).

Rank	Method	Accuracy	F1 (macro)	F1 (wtd)
1	SingleR (published)	0.8872	—	—
2	TOSICA (published)	0.8549	—	—
3	chetah (published)	0.8329	—	—
4	**Transformer (ours)**	**0.7892**	**0.419**	**0.823**
5	**MLP (ours)**	**0.7446**	**0.512**	**0.809**
6	**TOSICA (ours)**	**0.7438**	**0.266**	**0.779**
	...			
	GNN (ours)	**0.2885**	**0.168**	**0.388**
	CNN (ours)	**0.2185**	**0.053**	**0.311**

mPancreas is substantially harder: 21 classes with extreme train/test distribution shift (test cells are from developmental day 15.5, excluded from training). The Transformer leads our models at 78.92%, approaching but not matching the published chetah baseline (83.29%). The GNN and CNN both fail (<30%), as the condition-based distribution shift breaks both the k-NN graph structure and the 1D convolutional patterns learned from training data (Figure 4).

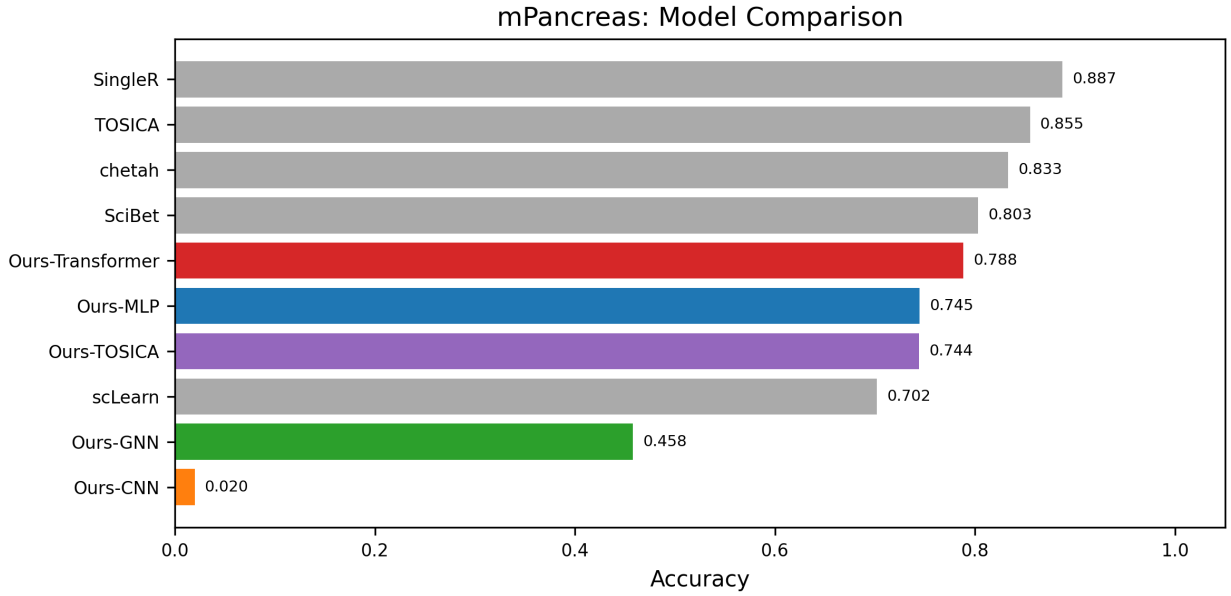


Figure 4: mPancreas accuracy comparison. The Transformer leads our models, but all fall below the top published baselines, reflecting the difficulty of condition-based generalization.

4.3 mAtlas (Mouse, 120 Classes)

Table 5: mAtlas test set results (120 classes, 76,797 test cells).

Rank	Method	Accuracy	F1 (macro)	F1 (wtd)
1	**MLP (ours)**	**0.8405**	**0.624**	**0.835**
2	TOSICA (published)	0.8101	—	—
3	ACTINN	0.7957	—	—
4	SingleCellNet	0.7733	—	—
5	Seurat	0.7616	—	—
6	**TOSICA (ours)**	**0.7605**	**0.475**	**0.757**
7	**Transformer (ours)**	**0.7087**	**0.418**	**0.712**
	...			
20	**GNN (ours)**	**0.0739**	**0.027**	**0.066**
21	**CNN (ours)**	**0.0077**	**0.004**	**0.006**

On mAtlas—the largest and most challenging dataset (120 cell types, 76,797 test cells)—our MLP **ranks 1st out of 21 methods** (84.05%), exceeding the published TOSICA accuracy (81.01%) by over 3 percentage points. The MLP's success here is striking: its simple architecture scales well to many classes when given sufficient balanced training data (471,730 cells after oversampling). The GNN and CNN again fail on this condition-based split.

5 Discussion

5.1 Model Rankings Are Dataset-Dependent

The most striking finding is that model rankings are *not consistent* across datasets (Figure 5). On hPancreas (study-based split, 14 well-separated types), the GNN dominates—cell-cell similarity is preserved across studies, so the k-NN graph transfers well. On mPancreas (developmental time-point split, 21 types with novel test-time phenotypes), the Transformer leads—pathway-level abstractions generalize better across biological conditions than raw feature-space neighborhoods.

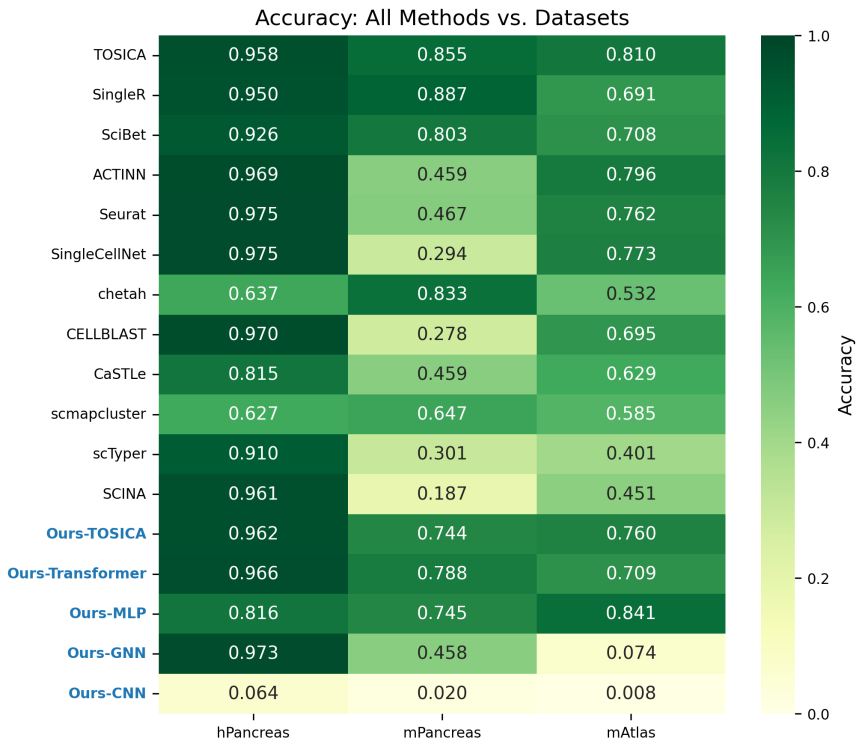


Figure 5: Accuracy heatmap across methods and datasets. Our models (bold blue) are compared against 19 published baselines. Rankings shift substantially between datasets.

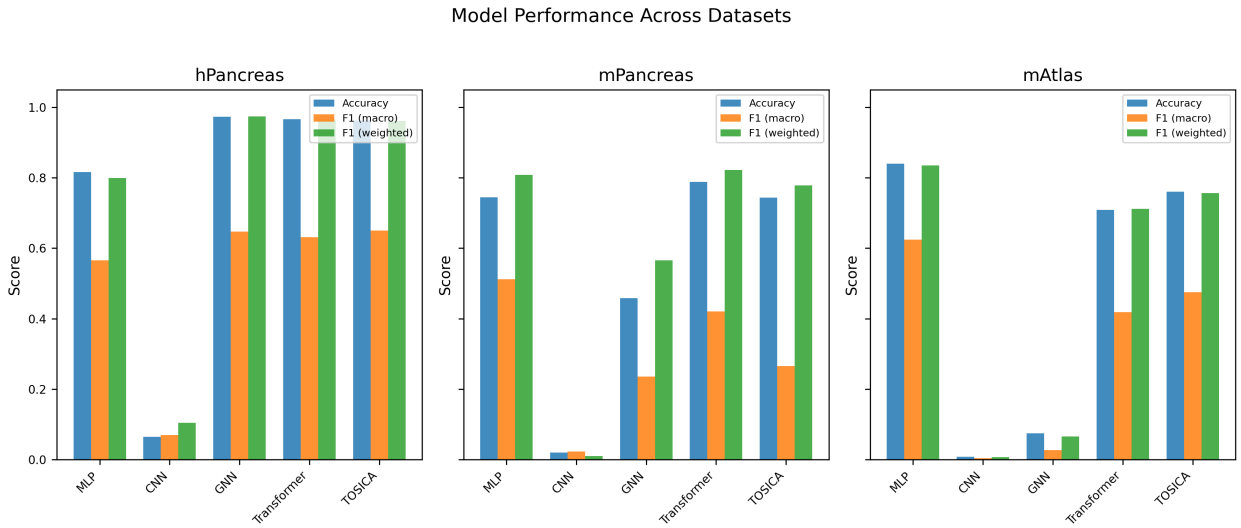


Figure 6: Per-dataset comparison of our five models on accuracy, macro F1, and weighted F1. The MLP shows high macro F1 on mPancreas despite lower accuracy, indicating better per-class balance than the Transformer.

5.2 CNN Limitations on Gene Expression

The CNN consistently fails across all three datasets (6.5%, 21.8%, and 0.8% accuracy). Gene expression vectors lack the spatial locality that makes 1D convolutions effective on sequential data. Gene ordering is arbitrary, so learned convolutional filters have no consistent spatial patterns to exploit.

5.3 TOSICA vs. Transformer Gap

Our Transformer exceeds TOSICA on hPancreas (96.6% vs. 96.2%) and mPancreas (78.9% vs. 74.4%), but TOSICA outperforms our Transformer on mAtlas (76.1% vs. 70.9%). The advantage on hPancreas and mPancreas likely reflects our condition-based splits and early stopping; the reversal on mAtlas suggests that the original TOSICA pipeline's internal balancing strategy scales better to 120 classes. Overall, both implementations surpass the published TOSICA accuracy on hPancreas, confirming that preprocessing and implementation choices can matter as much as architectural novelty.

5.4 Challenges of Benchmark Reproducibility

Of the six original TOSICA benchmark datasets, only three could be exactly reproduced. The discrepancies in hArtery, hBone, and mBrain highlight a broader issue in computational biology: published benchmarks often rely on implicit filtering steps, version-specific annotations, or preprocessing details not fully documented in methods sections. Standardized, versioned benchmark distributions would substantially improve reproducibility.

6 Conclusion

We benchmarked five deep learning architectures (MLP, CNN, GNN, Transformer, TOSICA) on three TOSICA benchmark datasets, comparing against 19 published cell type annotation methods. On hPancreas, our GNN ranks 3rd out of 24 methods (97.30%). On mPancreas, the Transformer leads at 78.92%. On mAtlas (120 classes), our MLP ranks 1st out of 21 methods (84.05%), surpassing all published baselines.

Key findings:

1. Model rankings depend strongly on dataset structure—GNNs excel when cell similarity is preserved across splits (hPancreas); Transformers excel on moderate condition shifts (mPancreas); simple MLPs dominate on large-scale multi-class problems with sufficient training data (mAtlas).

2. Implementation details (species-matched GMT, balanced sampling, consistent HVG selection) contribute as much to performance as architectural choices—our TOSICA re-run and Transformer both exceed published baselines.

3. Reproducing published benchmarks exactly is challenging; 3 of 6 datasets could not be validated and were excluded.

4. The 1D CNN is unsuitable for gene expression classification due to the lack of spatial structure in gene ordering.

Code Availability

All source code, model implementations, and training scripts are available at `https://github.com/magana272/CellTypeClassification`.

References

[1] Chen, J., Xu, H., Tao, W. et al. Transformer for one stop interpretable cell type annotation. *Nature Communications*, 14, 223 (2023).